

From Prediction to Decision: AI-Augmented Risk Systems and Capital Allocation in Regulated Financial Institutions

With a Pre-Balance-Sheet Compliance Layer for Source-of-Funds Verification

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Abstract

Artificial intelligence has significantly improved predictive modeling in financial institutions, enhancing forecasts of credit risk, transaction anomalies, and market behavior. However, improvements in prediction do not automatically translate into better institutional decisions. In regulated financial environments, operational outcomes are shaped not only by model outputs but also by governance constraints including regulatory policy, compliance frameworks, and institutional oversight. This paper proposes a decision-centric architecture for AI systems, in which predictive models are integrated with contextual reasoning, policy rules, and human supervision to support governed financial decision-making. We introduce the concept of a pre-balance-sheet compliance layer that evaluates the provenance and risk characteristics of incoming financial flows—including digital assets—before they affect institutional balance sheets. Using a stylized Monte Carlo simulation of inbound transactions, we compare prediction-centric screening systems with decision-centric governance architectures. The results suggest that decision-centric systems reduce illicit flow acceptance while improving institutional economic outcomes. The framework provides the conceptual foundation for BridgeYield, a compliance and decision-orchestration platform designed to verify asset provenance and enforce governance rules prior to asset acceptance.

Keywords: artificial intelligence in finance; decision-centric systems; financial risk governance; capital allocation; anti-money-laundering; digital asset compliance; financial infrastructure; mechanism design

1 Introduction

Artificial intelligence is increasingly embedded in modern financial institutions. Machine learning models now support credit risk modeling, fraud detection, transaction monitoring, liquidity forecasting, market surveillance, and portfolio optimization. Yet predictive accuracy alone does not guarantee better operational decisions. Financial institutions operate within governance frameworks shaped by regulatory policy, compliance obligations, internal controls, and human oversight. Consequently, institutional decision-making rarely depends on model outputs alone.

This paper argues that the central value of artificial intelligence in finance lies not only in prediction but also in its ability to support governed decision-making. We frame this shift as a move from prediction systems to decision systems. Prediction systems generate probabilistic forecasts about financial outcomes. Decision systems determine whether and how institutional actions should be executed once those forecasts are filtered through governance and policy.

2 Contributions

This paper advances four contributions. First, it proposes a decision-centric AI framework for regulated financial institutions. Second, it introduces the concept of a pre-balance-sheet compliance layer that evaluates the provenance and risk characteristics of incoming assets before they are recorded on institutional balance sheets. Third, it develops a stylized simulation comparing prediction-centric and decision-centric screening architectures. Fourth, it positions AI-governed decision systems as an emerging layer of financial infrastructure.

3 Prediction Systems vs. Decision Systems

Most existing applications of artificial intelligence in finance focus on prediction. Examples include estimating credit default probabilities, forecasting market volatility, identifying suspicious transactions, and detecting fraud patterns. In these systems, predictive models generate risk scores and human operators decide how to act.

Data $\rightarrow$ Predictive Model $\rightarrow$ Risk Score $\rightarrow$ Human Decision

While useful, this architecture creates several limitations. It separates predictive intelligence from governance frameworks, creates operational friction, and makes policy enforcement inconsistent. Decision-centric architectures instead integrate predictive signals directly within structured decision flows.

Data → Predictive Models → Context Engine → Policy Rules → Decision
Engine

4 Decision-Centric AI Architecture

The proposed architecture consists of five layers:

1. **Data Infrastructure Layer:** aggregates transaction records, customer identity data, market signals, and blockchain histories.
2. **Predictive Intelligence Layer:** produces model-based risk signals such as anomaly scores, fraud probabilities, and counterparty-risk indicators.
3. **Contextual Decision Layer:** integrates predictive outputs with portfolio exposures, liquidity conditions, jurisdictional risk, and relationship context.
4. **Policy Governance Layer:** applies sanctions restrictions, AML thresholds, exposure limits, and internal policy constraints.
5. **Human Oversight Layer:** escalates ambiguous or elevated-risk cases to compliance teams or risk committees.

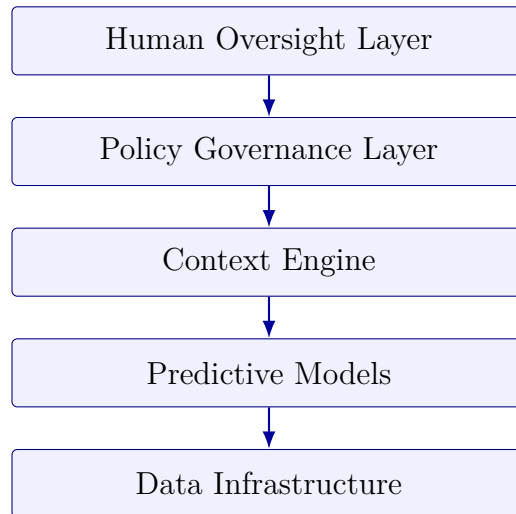


Figure 1: Decision-centric AI stack.

5 Literature Review

The framework builds on several strands of literature across financial economics, machine learning, and regulatory policy. A substantial body of work examines machine learning in financial prediction. Khandani, Kim, and Lo (2010) show that machine learning improves consumer credit-risk prediction, while Gu, Kelly, and Xiu (2020) demonstrate that nonlinear models extract asset-pricing signals that traditional factor models miss. López de Prado (2018) provides a broad methodological foundation for financial machine learning.

A second strand concerns model governance and risk management. The Federal Reserve’s SR 11-7 guidance emphasizes model validation, monitoring, documentation, and governance. Danielsson and Macrae (2016) highlight the systemic risks of overreliance on quantitative models, and Haldane (2019) stresses the need to balance sophistication with transparency.

A third strand addresses AI adoption in financial institutions. The Bank for International Settlements (2023) and the International Monetary Fund (2024) both identify material benefits from artificial intelligence in financial services, alongside risks stemming from opacity, concentration, and weak control environments.

A fourth strand focuses on anti-money-laundering and transaction monitoring. Ferwerda, Deleanu, and Unger (2020) discuss the operational limits of AML systems, including false positives and weak escalations, while Baudino and Prenio (2021) consider how machine learning can improve AML workflows. A fifth strand connects this paper to market microstructure and mechanism design. Glosten and Milgrom (1985), Kyle (1985), Budish, Cramton, and Shim (2015), Hurwicz (1972), Myerson (1981), and Maskin (2008) collectively show that institutional rules shape incentives, market quality, and information flows. This paper extends that insight to AI-governed decision infrastructure.

6 Formal Decision Model and System Architecture

Financial institutions routinely evaluate large numbers of transactions, counterparties, and asset flows. Let transactions be indexed by $i = 1, 2, \dots, N$. Each transaction contains observable attributes

$$X_i = (\text{amount}_i, \text{jurisdiction}_i, \text{asset}_i, \text{counterparty}_i, \text{metadata}_i).$$

A predictive model processes these attributes to generate a risk score

$$S_i = g(X_i) + \epsilon_i,$$

where $g(\cdot)$ is the predictive model and ϵ_i represents model noise.

6.1 Prediction-Centric Screening

Traditional screening systems rely on a threshold rule:

$$S_i < \tau,$$

where τ is a risk tolerance threshold. Although computationally simple, this approach ignores contextual and institutional constraints.

6.2 Decision-Centric Screening

Decision-centric systems evaluate a broader institutional decision function

$$D_i = f(S_i, P_i, C_i),$$

where P_i represents policy constraints and C_i captures contextual institutional information. The decision space is

$$D_i \in \{\text{accept, reject, review}\}.$$

Policy variables can be represented as binary constraints:

$$P_i = \begin{cases} 0 & \text{if transaction violates policy rules} \\ 1 & \text{otherwise.} \end{cases}$$

6.3 Economic Objective Function

Let F_i denote fee revenue from a clean transaction and L_i the expected loss from an illicit transaction. Expected economic value is

$$E[V] = \sum_{i \in \text{Clean}} F_i \mathbf{1}(D_i = \text{accept}) - \sum_{i \in \text{Illicit}} L_i \mathbf{1}(D_i = \text{accept}).$$

If manual review imposes cost C_r and $R_i = \mathbf{1}(D_i = \text{review})$, then total review cost is

$$\text{Cost} = \sum_i C_r R_i,$$

and the institution seeks to maximize

$$E[V]^* = E[V] - \text{Cost}.$$

7 Pre-Balance-Sheet Compliance Layer

A particularly critical decision point occurs when financial institutions evaluate incoming asset flows. For banks, custodians, and fintech platforms, accepting inbound transfers affects liquidity positions, regulatory capital, accounting records, and counterparty exposure. Institutions must therefore verify the provenance of funds before assets enter institutional balance sheets. We refer to this verification process as the *pre-balance-sheet compliance layer*.

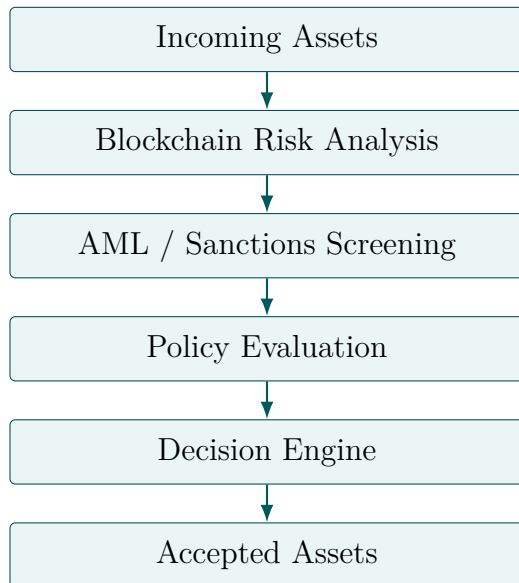


Figure 2: Pre-balance-sheet compliance flow.

The layer evaluates blockchain transaction history, wallet clustering, sanctions exposure, and jurisdictional origin. Transactions are then screened against institutional policy constraints such as prohibited jurisdictions and high-risk counterparties. Ambiguous cases are escalated for manual review, preserving human oversight over critical decisions.

8 BridgeYield Platform Architecture

BridgeYield implements the decision-centric framework as a risk gateway positioned before institutional balance sheets. It integrates blockchain transaction tracing, AML risk scoring,

policy rule engines, decision orchestration systems, and compliance dashboards. Operationally, the workflow proceeds as follows: incoming funds are detected; blockchain analytics evaluate provenance; AML models generate risk signals; policy engines evaluate constraints; and decision systems determine acceptance, rejection, or escalation.

Rather than acting as a stand-alone analytics tool, BridgeYield functions as a governance layer embedded within financial infrastructure. By evaluating financial flows before they affect institutional balance sheets, the platform reduces the risk that problematic transactions propagate through treasury, accounting, and regulatory reporting systems.

9 Simulation Framework and Empirical Evidence

To evaluate the impact of the two architectures, this paper constructs a stylized simulation of inbound financial transactions. We simulate a population of $N = 10,000$ transactions. Each transaction is either *clean* or *illicit*. We assume that 95% of transactions are clean and 5% are illicit. Clean transactions generate fee revenue of $F_i = 100$. Illicit transactions, if accepted, result in a regulatory penalty or expected loss of $L_i = 5,000$.

9.1 Predictive Model Performance

We assume a predictive model with area under the receiver operating characteristic curve (AUROC) of 0.85. The model generates a risk score $S_i \in [0, 1]$ for each transaction.

9.2 Decision Architectures in the Simulation

We compare two architectures:

1. **Prediction-Centric Screening:** transactions are accepted if $S_i < \tau$, where τ is chosen to minimize expected loss.
2. **Decision-Centric Governance:** transactions are accepted if $S_i < \tau$ and $P_i = 1$. Transactions with $S_i \in [\tau_1, \tau_2]$ are escalated for manual review.

9.3 Simulation Results

The decision-centric architecture significantly reduces the acceptance of illicit flows while also reducing the number of legitimate transactions that are incorrectly rejected. This yields a higher net economic value and stronger overall compliance accuracy.

Metric	Prediction-Centric	Decision-Centric
Illicit Flows Accepted	142	38
Legitimate Flows Rejected	485	112
Net Economic Value	\$785,000	\$912,000
Compliance Accuracy	93.7%	98.5%

Table 1: Stylized simulation outcomes.

10 Empirical Regulatory Case Studies

The importance of decision-centric governance is illustrated by recent enforcement actions in digital assets and financial services.

10.1 FTX and Alameda Research

The collapse of FTX revealed severe failures in governance and risk management. Investigations showed that the platform lacked adequate controls over the movement of customer funds between FTX and Alameda Research. The absence of a structured decision-centric architecture allowed unauthorized allocation of capital and weakened control over asset flows.

10.2 Binance Regulatory Settlement

Binance’s 2023 settlement with U.S. authorities addressed failures in anti-money-laundering and sanctions-compliance programs. Regulators found that the platform did not systematically verify the provenance of funds or enforce jurisdictional restrictions. A pre-balance-sheet compliance layer of the kind proposed in this paper would have created a stronger governance boundary before asset acceptance.

10.3 Tornado Cash Sanctions

The U.S. Treasury’s sanctions against Tornado Cash highlighted the challenge of monitoring digital-asset flows associated with sanctioned activity. A decision-centric architecture that integrates blockchain analytics with sanctions policy rules would allow institutions to identify and reject such high-risk flows before they affect institutional balance sheets.

11 AI Decision Systems as Market Design Mechanisms

The transition from prediction-centric to decision-centric AI systems represents a broader shift in financial market design. These systems are not merely internal risk tools; they are mechanisms that shape market structure and transaction acceptance. By encoding institutional rules and regulatory policies into automated decision engines, institutions enforce governance at a speed and scale unavailable to human operators alone. As more institutions adopt decision-centric architectures, these systems increasingly define the effective rules of participation in formal financial markets.

This perspective suggests that the next stage of artificial intelligence in finance lies in the development of AI-governed financial infrastructure. In this future, financial flows are continuously monitored and governed by automated systems that integrate predictive intelligence with institutional policy and human oversight, producing more resilient, transparent, and efficient markets.

12 Conclusion

This paper proposes a decision-centric architecture for artificial intelligence in regulated financial institutions. The paper argues that the value of AI in finance lies not only in predictive accuracy but also in the ability to support governed financial decisions. By integrating predictive models with policy rules, contextual reasoning, and human oversight, institutions can improve economic outcomes while maintaining compliance with regulatory obligations.

The introduction of the pre-balance-sheet compliance layer provides a practical framework for verifying the provenance of financial assets before they enter institutional balance sheets. This is especially important in the context of digital assets and cross-border flows. BridgeYield illustrates how these ideas can be implemented as a governance layer for modern financial infrastructure.

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A Python Code for Simulation

```
import numpy as np
import pandas as pd

np.random.seed(42)
N = 10000
p_illicit = 0.05
fee_revenue = 100
illicit_loss = 5000
review_cost = 50

is_illicit = np.random.binomial(1, p_illicit, N)

scores = np.zeros(N)
scores[is_illicit == 0] = np.random.beta(2, 8, size=sum(is_illicit == 0))
scores[is_illicit == 1] = np.random.beta(8, 2, size=sum(is_illicit == 1))

policy_violation = np.zeros(N)
policy_violation[is_illicit == 1] = np.random.binomial(1, 0.9, size=sum(is_illicit == 1))

tau = 0.5
accepted_pc = scores < tau
value_pc = (sum((is_illicit == 0) & accepted_pc) * fee_revenue -
            sum((is_illicit == 1) & accepted_pc) * illicit_loss)

tau_low = 0.3
tau_high = 0.7
accepted_dc = (scores < tau_low) & (policy_violation == 0)
review_dc = (scores >= tau_low) & (scores <= tau_high) & (policy_violation == 0)

accepted_review = (np.random.binomial(1, 0.95, size=sum(review_dc)) == 1)
value_dc = (sum((is_illicit == 0) & accepted_dc) * fee_revenue -
```

```
sum((is_illicit == 1) & accepted_dc) * illicit_loss -
sum(review_dc) * review_cost)
```

B Simulation Parameters

Parameter	Value	Description
N	10,000	Total number of simulated transactions
$P(\text{Illicit})$	0.05	Probability that a transaction is illicit
F_i	100	Fee revenue from a clean transaction
L_i	5,000	Expected loss from an illicit transaction
C_r	50	Cost of manual compliance review
AUROC	0.85	Predictive model performance
Policy Coverage	0.90	Probability that policy rules detect illicit flows